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JULY–SEPTEMBER, 2025



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### Dus Gajhat – the deaths of ten elephants

Conflict often becomes inevitable in landscapes where livelihoods depend on the same land that sustains wildlife.

**Gobind Sagar Bhardwaj** and **Ramesh Kumar Pandey** recount the story of ten elephants from Bandhavgarh Tiger Reserve whose deaths sent shockwaves through the forest department, conservationists, and policymakers alike. The official conclusion pointed to food poisoning caused by infected kodo millet. Yet, the Bandhavgarh tragedy is more than a story of poisoned elephants – it stands as a stark reminder of the fragile coexistence between people and nature. Read on.



### Birding Diaries

A walk in the forest is nothing short of meditation for **Shalini Gopalakrishnan**. Her birding expeditions have taken her to many remote corners of our beautiful country, and along the way, one truth has stood out for her: in the avian world, one encounters some of nature's finest singers, dancers, actors, architects, and acrobats. With tongue-twisting names, dazzling colours, and a spirit of camaraderie, these birds embody simplicity, humility, and grace. In this piece, she shares her experiences exploring the avian wonders of the Darjeeling and Kalimpong districts of West Bengal.

## Others

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# Editorial...

Hope you had a wonderful festive season! Diwali, the festival of lights, reminds us of hope and renewal – and I am sure *Hornbill* will continue to shine as a beacon of hope for conservation. Warm greetings to you on this auspicious occasion!

At the outset, let me thank you for sharing your feedback on the fascinating article “*Crow: Intelligent Villain or Misunderstood Genius?*” by Dr Asad R. Rahmani in the previous issue. If we were to publish all your observations together, they would make for another excellent piece!

It is truly remarkable that India – a country with a population density of 492 people per square kilometre – has still managed to protect its ‘Big Five’: the tiger, Asiatic lion, leopard, elephant, and one-horned rhinoceros. These species are not easy to conserve. They require optimal and unique habitats and connectivity through wildlife corridors. At a time when humans are fighting for every inch of land, it becomes increasingly difficult for voiceless species to find space.

Elephants, being intelligent and revered as deities in Indian culture, have managed to survive against all odds – and are even reclaiming parts of their historical range. This journey has been far from easy and has involved overcoming numerous challenges. A few months ago, Indian media reported a tragic incident – the deaths of ten elephants in the Bandhavgarh Tiger Reserve. Two Indian Forest Service officers, Gobind Sagar Bhardwaj and Ramesh Kumar Pandey, have written about on this incident, with official investigations pointing to food poisoning caused by infected Kodo millet. They believe that the Bandhavgarh tragedy goes beyond poisoned elephants – it stands as a stark reminder of the fragile coexistence between humans and nature.

Shalini Gopalakrishnan has been birding across remote corners of the country for years. Through her experiences, she has encountered some of nature’s finest singers, dancers, architects, and acrobats. The tongue-twisting names, dazzling colours, and effortless grace of birds, for her, embody simplicity and camaraderie. A long-held wish to explore the avian wonders of West Bengal’s Darjeeling and Kalimpong districts was fulfilled through a BNHS Programmes camp. Latpanchar, Lava, and Neora Valley National Park became her home for a week. In this piece, she shares memories from a journey filled with binoculars, cameras, and boundless enthusiasm.



This issue features four wonderful books, including one for children. *THE SEARCH FOR INDIA’S RAREST BIRDS*, edited by Shashank Dalvi and Anita Mani (Juggernaut Books, 2024); *BIRDS OF INDIA – THE NEW SYNOPSIS* by Praveen J. (Nature Conservation Foundation, 2025); *MOTHS OF AGASTYA* by R. Bhanumathi (Agastya International Foundation, 2025); and *A RACE OF THE TURTLES* by Abhishek Jakhade (Padmagandha





Prakashan). These are reviewed by Parveen Shaikh, Harkirat Singh Sangha, Kumaran Sathasivam, and Priyadarshini Supekar respectively. We hope you enjoy reading them!

In the ‘Builders and Guardians’ section, you will read about William Mansfield, Lord Sandhurst – the 3rd President of BNHS, who served from 1895 to 1900. Lord Sandhurst, also known as Lord William Barron Sandhurst, GCIE, was the Governor of the Bombay Presidency during a difficult time marked by the bubonic plague outbreak in 1896, which claimed over 20,000 lives and caused mass exodus from the city.

Field Notes of a Naturalist section, meet Khushboo Rani, a Project Associate at BNHS, who shares her experiences from the Vikramshila Gangetic Dolphin Sanctuary (VGDS) in Bihar through *Islands of Hope*. Khushboo’s passion for the Ganges sandbars and her extensive experience in bird migration and wetland ecosystem studies offer valuable insights.

Nature is full of discoveries! One such rare sighting featured in ‘Readers’ Space’ shows Indian cormorants *Phalacrocorax fuscicollis* hunting bullfrogs in wetlands near Lal Dhoori, nestled at the foothills of Girnar in the Bhavnath area of Junagadh, Gujarat.

In the ‘Nature Watch’ section, wildlife enthusiast and coral restoration advocate Dhrishit Khandhar contributes a fascinating piece on the corals of the Andaman and Nicobar Islands. He recounts a ferry trip to Swaraj Dweep (formerly Havelock Island) and underwater observations during dives at Kalapatthar Beach and Inchkeith Wreck near Havelock.

Rupjyoti Rabha and Sachin Ranade share noteworthy nesting observations of the Purple-rumped Sunbird *Leptocoma zeylonica* at Goalpara. The findings are significant because Goalpara lies just north of the species’ usual range, and because the birds used anthropogenic materials such as plastic for nest-building – a growing global trend among more than 6,000 bird species worldwide.

While most nature lovers visit the Sundarbans to catch a glimpse of the tiger, Subir Ghosh draws attention to the fascinating amphibious mudskippers – remarkable fish capable of climbing, walking, and even jumping on land.

BNHS has achieved much in its 140+ years of conservation work, yet grasslands remain among the least understood and least protected ecosystems in India. Readers may be interested to know that BNHS has previously worked with the Government of India to provide specific recommendations for grassland conservation. After reading *Why Are Grasslands Ignored in India?* you may feel, as we do, that more must be done to convince policymakers to act urgently. We hope this issue serves as a catalyst for change, and we urge readers to write to the MoEF&CC and their respective state authorities, advocating for the scientific protection and management of grasslands.

Enjoy reading this issue!

Kishor Rithe



# Dus Gajhat –



Text: **Gobind Sagar Bhardwaj and Ramesh Kumar Pandey**  
Photographs: **Gobind Sagar Bhardwaj**

*A strong stench of decay hung heavy in the air near board number ten – a small wooden sign marking the burial site of the tenth elephant from a herd of thirteen. A lone golden jackal circled the resting place of the eighth elephant, buried some fifty metres away, nose twitching, drawn by the scent of rotting flesh. When it noticed us, it bolted toward the forest, vanishing into the sal trees. The pungent odour of rotting flesh must have attracted the scavenger, which darted back into the woods as soon as it sensed our presence.*

*Ten elephants lay buried here – victims of a tragedy that had shaken the forests of Bandhavgarh Tiger Reserve (BTR). Their sudden deaths under mysterious circumstances had sent shockwaves through the forest department, conservationists, and policymakers alike. The burial site, adjoining Salkhanja village in the buffer zone of the Bandhavgarh Tiger Reserve (BTR) – Dus Gajhat – had become the focus of national attention.*



# the deaths of ten elephants



**B**andhavgarh, in northeastern Madhya Pradesh, is best known for its tigers. Tourists from across the world come here for a chance to glimpse the striped king of the jungle. Until recently, elephants were never part of its identity. In fact, wild elephants had been absent from this region for decades.

That changed in 2018, when a scattered herd of about forty elephants wandered into the eastern edge of the reserve. Initially, officials assumed it was a temporary visit — perhaps a group exploring new grounds that would soon return home. The elephants stayed. They began

to settle, carving out a new home in these sal forests.

No one knows exactly why they migrated. Perhaps their traditional habitats had become fragmented by human expansion, or maybe changing climate patterns and increasing disturbances may have caused a population split, leading smaller groups to disperse and seek refuge in new landscapes such as BTR. Whatever the reason, Bandhavgarh became an unlikely refuge for elephants — until tragedy struck.

News of the alleged poisoning of ten elephants spread like wildfire. Theories circulated wildly: deliberate poisoning, tainted crops, disease. As



members of the national Project Elephant team, my colleague and I closely followed the unfolding developments. We soon decided to visit the site ourselves to collect field information, and interact directly with local officials and villagers.

Our journey began in Jabalpur, winding through villages, forests, and paddy fields toward Bandhavgarh. The reserve – with the tiger as its flagship species – is named after the ancient Bandhavgarh Fort and sprawls across more than 1,500 square kilometres between the Vindhyan and Satpura ranges. It is a breathtaking landscape, a living painting of valleys, hills, and plains. But beauty alone could not hide the unease that had gripped the forest. In recent years, BTR had already been under media scrutiny due to several cases of tiger mortality. Now, it was elephants, ten of them dead in one blow.

While travelling, we gathered as much information as possible. The Madhya Pradesh government had already formed an inquiry committee headed by Mr L. Krishnamurthy, APCCF, an experienced wildlife manager. Simultaneously, the Wildlife Crime Control Bureau (WCCB) of the central government had been tasked with investigating any possibility of unnatural deaths – whether through poaching or retaliatory killing. By then, preliminary field

reports suggested that the elephants had possibly died after consuming kodo millet, a locally grown grain, contaminated by a toxic fungus.

A forest jeep took us through a rough dirt track winding across the reserve's buffer zone. We passed half-relocated villages, surrounded by paddy fields fenced with brushwood. We soon entered a patch of the core area before crossing into the buffer forest dominated by sal *Shorea robusta* and its associated vegetation. The poor health of this buffer forest was striking – little ground vegetation, minimal understorey, and clear signs of illegal girdling – where bark is stripped to slowly kill trees without technically cutting them.

Finally, we reached the outskirts of Sankalia village in the Umaria Range, where the tragedy had unfolded. A uniformed forest guard, likely the local beat in-charge, greeted us and led us towards the actual field where the elephants had fed on *kodo* millet *Paspalum scrobiculatum* and, supposedly met their fate. His sombre expression and downcast eyes betrayed a sense of personal responsibility for the tragedy. He claimed that the elephants had eaten the kodo crop and that it was believed to have caused fatal poisoning. Most of the crop had already been harvested and burned to prevent further losses.



Panicles of Kodo millet





Girdling and lynching of trees to accelerate their death, without the use of an axe, prior to encroachment was observed as a modus operandi in the area

A wave of dismay swept through us as we realised the standing crop had been destroyed – vital evidence lost to fire. Fortunately, a few tufts of kodo millet plants still clung to life beneath trees and bushes, where the sickles could not reach. We examined them closely and spotted clusters of infected grains – samples of the infected millet were collected for analysis.

Kodo millet, a member of the grass family Poaceae, is one of India's oldest cereals, cultivated for over 3,000 years. Known locally as *kodo* in Hindi and *varagu* in Tamil, it was once a staple food in southern India. The crop is hardy, drought-resistant, and thrives on poor soils with minimal irrigation. It is a *kharif* crop, often grown by farmers in Madhya Pradesh, Chhattisgarh, and southern states as a fall-back when paddy fails.

Under normal circumstances, kodo is harmless. But when the crop gets infected by certain fungi – especially *Aspergillus flavus* and *A. tamarii* – it can become toxic. These fungi thrive in moist conditions, releasing a mycotoxin called Cyclopiazonic Acid (CPA). In humans and animals, CPA can damage the liver, heart, and nervous system, leading to loss of coordination, paralysis, and even death.

Cases of kodo poisoning in humans have been recorded since the late 19th century. In 1870, medical officer Norman Chevers in his manual MEDICAL JURISPRUDENCE FOR INDIA, documented that contaminated kodo caused severe illness, locally called *malona* in Madhya Pradesh. In 1922, the *Indian Medical Gazette* reported similar poisoning incidents in Shahjahanpur, Uttar Pradesh. However, animal cases are extremely rare – the only notable record dates back to 1934, when 14 elephants reportedly died in Tamil Nadu after eating infected kodo millet '*kiruku varagu*'.

At the suspected field, we met a frail woman sitting beside a borewell, staring blankly at the gathered officials. Her teenage son hovered nearby, uneasy and silent. She was visibly uneasy at the sight of government officials. After much persuasion, and encouragement from women members of the visiting team, she finally spoke.

Four years ago, she said, elephants had raided her paddy fields, destroying her crop. She never received compensation. This year, to protect her paddy, she had installed an electric fence and planted Kodo millet around it as a buffer – a common local practice. The millet, left unprotected, acted as a 'sacrifice crop' to deter elephants from breaking into the main field.



When GSB asked the Deputy Field Director about her claim, he explained that compensation for crop damage is processed through the revenue department, with forest officials only assessing and reporting losses. The bureaucratic delay, however, often leaves farmers bitter and alienated.

As we walked back through the buffer forest, the human imprint was everywhere – much of the cultivation here occurred on encroached forest land. Dead sal trees bore the telltale marks of girdling – bark stripped in neat rings, a tactic used before converting forest patches into farmland. Once a tree dies naturally, locals can claim its wood without facing legal consequences. This grim practice reflects the harsh reality of life on the forest fringe, where survival often blurs the line between legality and necessity.

The next day, we visited neighbouring fields. Interestingly, other kodo millet patches looked dry but healthy, with no visible infection. Only the field where the elephants fed had moisture-retaining soil and dense shade – perfect conditions for fungal growth. So, was this an unfortunate coincidence? Or something deliberate?

Scientific literature suggested that when rainfall coincides with the ripening of kodo millet, fungal infection increases dramatically. Still, the thought of deliberate contamination lingered uneasily in my mind.

Even assuming the fungus was natural, several questions haunted us: Why did the elephants eat only kodo millet when paddy – their usual favourite – grew nearby? Why were no cattle, deer, or even granivorous birds affected? How could ten massive elephants – creatures of remarkable intelligence – fail to detect the poison?

Elephants are incredibly intelligent animals, with brains weighing about 4.8 kilograms – three times that of humans. They are capable of memory, empathy, and even self-recognition. One would expect them to sense danger.

Yet, in this case, desperation might have overridden instinct. The herd, perhaps hungry and stressed, could have been deterred from the power-fenced paddy and turned to the easily accessible kodo millet. Once they began feeding, it would have been nearly impossible to separate infected grains from healthy ones. Unlike cattle, which can regurgitate or vomit, elephants are monogastric – their single-chambered stomachs cannot expel toxins once ingested. A few mouthfuls of infected grain, consumed quickly and in bulk, could have been fatal.

Villagers mentioned that people who accidentally eat bad kodo millet often treat themselves with buttermilk or tamarind water. Livestock, it seems, are rarely affected – perhaps because they graze on mixed fodder rather than a single contaminated crop.



(L-R): Nandkishor Kale, AIG, NTCA, Nagpur and Gobind Sagar Bhardwaj (the author) interacting with staff and villagers in the field



Subsequent toxicological tests by the Indian Veterinary Research Institute confirmed the presence of Cyclopiazonic Acid in the elephants' tissues. The official conclusion: food poisoning from infected Kodo millet.

But as a field biologist, GSB remained uneasy. GSB had seen elephants feed on hundreds of wild plant species – some mildly toxic – without issue. Never before had there been a documented case of such mass mortality from food poisoning. Could this particular variety of kodo be more susceptible to fungal infection? Was the toxin concentration unusually high? Or did prolonged stress and malnutrition make the elephants more vulnerable?

These questions demanded deeper clinical study – mycological cultures, quantitative toxin analysis, perhaps even genetic testing of the millet strains. But in the bureaucratic rhythm of wildlife management, once the official report was filed, the world moved on.

Locals began calling the incident *Das Gajhat* – the death of ten elephants. Even now, whenever we think of that name, a heaviness settles in my chest.

Who was truly responsible? The fungus? The crop? The woman who unknowingly planted it? The system that delayed compensation? Or us – the policymakers who still struggle to bridge the gap between human need and wildlife conservation?

Bandhavgarh's tragedy is more than a mystery of poisoned elephants. It is a mirror reflecting the fragile coexistence between people and nature. Farmers here live on the edge – literally and figuratively – between forest and field, survival

and legality. When livelihoods depend on land that also shelters wild animals, conflict becomes inevitable.

The sight of those mounds – each marking the grave of a majestic creature – is seared into my memory. We remember the silence of the forest that day, broken only by the distant call of a peacock. It felt like the land itself was mourning.

Elephants hold a sacred place in Indian culture – symbols of Lord Ganesha, embodiments of wisdom and strength. To see them fall like this, not by bullets or traps but by something as humble as a grain, is deeply unsettling. The incident at Bandhavgarh underscores how fragile our ecosystems have become, how small disturbances – a fence, a crop, a fungus – can ripple through the web of life with devastating consequences.

Even as the case fades from public memory, it leaves behind important lessons. We need stronger systems for compensating crop losses, better monitoring of human-wildlife conflict zones, and research into the ecological risks of traditional crops like Kodo millet in elephant habitats. Most importantly, we must rekindle empathy – for both the farmer struggling to feed her family and the elephant searching for food in a shrinking forest.

When we close our eyes, we can still smell the faint trace of decay near board number ten – a haunting reminder that coexistence, while beautiful in theory, demands constant care, humility, and understanding.

If we fail to learn from the ten who fell at Bandhavgarh, we may soon be mourning many more. 



**Gobind Sagar Bhardwaj**, a senior forest officer (IFS 1994), has served across India's protected areas, led Project Bustard, conducted avian research, and passionately pursues wildlife photography.



**Ramesh Kumar Pandey**, a senior forest officer (IFS 1996), has managed Dudhwa Tiger Reserve, led Project Elephant, served in WCCB, and passionately works for elephant conservation.



# Birding Diaries from the East

Text and Photographs: **Shalini Gopalakrishnan**



“Look deep into nature,  
and you will understand  
everything better,” remarked  
Albert Einstein.

For me, a walk in the forest is nothing short of meditation. Add birding to it, and it becomes the most rejuvenating experience imaginable. A long-cherished wish to explore the avian wonders of the Darjeeling and Kalimpong districts of West Bengal was finally fulfilled through a BNHS Programmes camp. Latpanchar in the Mahananda Wildlife Sanctuary, Lava, and Neora Valley National Park were to be my home for the next week. Armed with binoculars, cameras, and uncontainable enthusiasm, we set off from Bagdogra toward our first destination.

Verditer Flycatcher at Neora Valley National Park





Rufous-necked Hornbill (female & male) at Latpanchar

### In Search of the ‘Gardeners of the Forest’

At Latpanchar, after depositing our bags at the homestay and a quick lunch, we set out to meet the celebrity residents of the region. The trail was as captivating as its inhabitants – rhododendrons were in full bloom, and a Malayan giant squirrel, comfortably perched on a branch, was busy foraging on fruits.

After trudging through the forest, we found ourselves stationed beneath a tall tree, necks craned, eyes fixed on a pair in courtship. Our search had ended. A rufous-necked hornbill was delicately plucking fruits for his mate, who accepted each offering before moving to a nearby branch to relish it. This ritual continued for a while.

Majestic and endearing, hornbills often mate for life. The male’s rufous head, neck, and underparts contrast beautifully with the female’s glossy black plumage. Being primarily frugivorous, hornbills play a vital role in dispersing seeds across the forest, earning them the affectionate title ‘Gardeners of the Forest’.

Some other birds spotted along the trail included barred cuckoo dove, sultan tit, red-tailed minla, scarlet and long-tailed minivet, and pale blue flycatcher.



Pale Blue Flycatcher at Latpanchar





Red-headed Trogon at Latpanchar



Blyth's Leaf Warbler at Lava Valley

### In Search of the 'Phantom of the East'

The grey-bellied tesia, a secretive little bird that prefers dense undergrowth, tested every ounce of our patience. It darted in and out of shadows, vanishing before we could get a proper glimpse – truly living up to its elusive reputation.

Nearby, a hoary-bellied squirrel was enjoying its breakfast when our guide suddenly whispered, "Trogon, Trogon"! That was the moment we had been waiting for. We rushed to the indicated spot, and there, perched silently amidst the mossy foliage, was the red-headed trogon – the 'Phantom of the East'. Still and regal, it surveyed the forest before swooping down to catch its prey, then pausing briefly near us, its vivid colours glinting in the dappled light. Within seconds, it vanished as quietly as it had appeared, leaving us spellbound.

Some other birds seen along the trail were long-tailed broadbill, common woodshrike, rufous-bellied niltava, Himalayan black-lored tit, bay woodpecker, and chestnut-bellied nuthatch.

### A Day of the Fast and the Feisty

I have always been fascinated by the speed and agility of babblers and warblers. Photographing them is an exercise in patience – the camera often chooses to focus on leaves, twigs, or anything except the bird that is merrily fluttering just out of reach.

The trail, lined with tall trees and low shrubs, provided perfect cover for a variety of species. A golden babbler pair was busily constructing their nest, darting to and fro with blades of tender grass – a scene as rapid and rhythmic as a tennis rally.

Soon after, an elegant white-browed shrike-babbler caught our eye, followed by chestnut-crowned laughing thrushes having a lively bath in a small stream. A cheerful red-billed leiothrix joined in, while a flock of rusty-fronted barwings descended on a nearby fruiting tree, swinging and fluttering until every berry was gone.

Other birds seen during the trail were green-backed tit, black-throated tit, and mountain tailorbird.



Slender-billed Scimitar-babbler at Lava Valley



White-browed Shrike-babbler at Neora

### Meeting the Bright, Spirited, Sunlit Beauties

Neora Valley is home to several varieties of my favourite birds – the sunbirds. The first we encountered was a rarity; Mrs Gould's sunbird. Tiny yet dazzling, it bounced among the treetops like a burst of living confetti. Those lucky enough to capture it through their lenses were left awestruck. Amid the greens and browns of the forest, it was a vivid explosion of colour.

Next came the green-tailed sunbird, iridescent and shimmering in the sunlight. And then, as if conjured by magic, a streak of red flashed overhead – the magnificent fire-tailed sunbird. Its elongated crimson tail feathers fluttered like a flag in the breeze, leaving us breathless with wonder.

Later that evening, we met yet another sunbird – subtle but equally enchanting. The black-throated sunbird, with its understated yet bold hues, was a fitting finale to a day filled with light and colour.

Merry, radiant, and truly sun-kissed, these birds carry the glow of the morning within them.



Fire-tailed Sunbird at Rishyap, Neora





Rufous-gorgeted Flycatcher at Neora Valley National Park



Red-headed Bullfinch at Rishyap, Neora

### In Search of the Satyr Tragopan

Our final day was devoted to tracking one of the most sought-after Himalayan pheasants – the satyr tragopan. With its black face, blue cheeks, greyish-brown back, and crimson plumage, it is a vision of beauty. The forest seemed straight out of ‘The Lord of the Rings’ – towering trees draped in moss, tendrils of mist curling through the branches, and an air of enchantment. We began our trek full of anticipation.

A Blyth’s leaf warbler flitted among the leaves, singing cheerfully. Nearby, a flock of Tibetan serin gorged on berries. As we climbed higher, we reached a ridge overlooking the ancient trade path once used to cross into Bhutan. There, from the dense undergrowth, we heard the familiar, piping call of the chestnut-headed tesia – another master of hide-and-seek. Despite our efforts, the elusive tragopan kept its mystery intact. With mixed feelings of awe and gratitude, we began our descent back to our homestay.

Other birds seen during the trail were rufous-gorgeted flycatcher, Nepal fulvetta, whiskered yuhina, red-faced liocichla, slender-billed scimitar babbler, black-crowned scimitar babbler, maroon oriole, and Himalayan cutia.

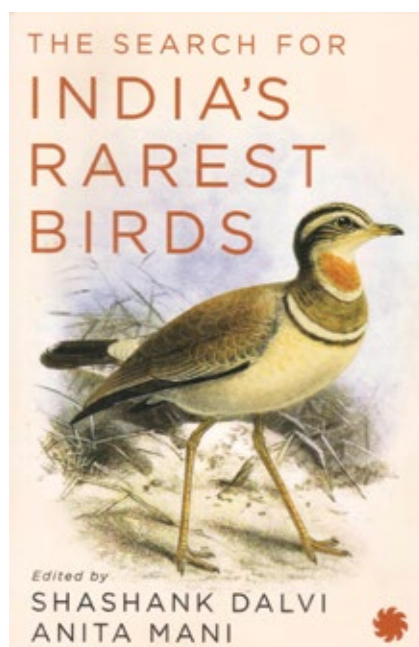
### Reflections

My birding journeys have taken me to many remote corners of our beautiful country. Over the years, one truth has stood out: when you enter the avian world, you encounter some of the finest singers, dancers, actors, architects, and acrobats of nature.

They bear tongue-twisting names, exhibit collaboration and camaraderie, and flaunt breathtaking colours and patterns – all with simplicity, humility, and grace. To truly appreciate them, one must experience them – not just see, but feel the music of their world. 🐦



**Shalini Gopalakrishnan** is an IT professional and a wildlife enthusiast and photographer by passion. She documents the natural world through words and images, hoping to inspire love and awareness for India’s unique biodiversity.



### **The Search for India's Rarest Birds**

Edited by: Shashank Dalvi and Anita Mani

Published by: Juggernaut Books, 2024

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Pages: 266

Price: ₹ 499/-

Paperback

Reviewed by: **Parveen Shaikh**

THE SEARCH FOR INDIA'S RAREST BIRDS is a meticulously researched and richly detailed account of the country's most elusive avian species and the explorers who brought them back from the brink of obscurity. Blending history, field biology, and adventure, it traces nearly two centuries of ornithological exploration – from colonial naturalists like A.O. Hume and T.C. Jerdon to contemporary Indian birders and scientists who are reclaiming their natural heritage through rediscovery and rigorous fieldwork.

The book documents both rediscoveries and new species found in post-independence India. Although only a handful of species have been described since 1947, each carries a remarkable story: the Mishmi wren-babbler (S. Dillon Ripley 1948), the Nicobar scops owl (Pamela Rasmussen 1998), the bugun liocichla (Ramana Athreya 2006), the Himalayan

forest thrush (Per Alström and Shashank Dalvi 2016), and the Ashambu sholakili (V.V. Robin 2017). These narratives reveal how years of observation, serendipity, and careful study of calls, habitats, and even DNA confirmed the existence of these species.

At its core, the book argues that rediscovery is not a matter of luck but of persistence. Pamela Rasmussen's rediscovery of the forest owlet exemplifies this spirit of perseverance – a methodical detective story built on museum research, ecological inference, and patient fieldwork. Likewise, Bharat Bhushan's search for Jerdon's courser, a ground-dwelling bird once believed extinct, captures this ethos perfectly. As the authors note, "the harder you work, the luckier you get".

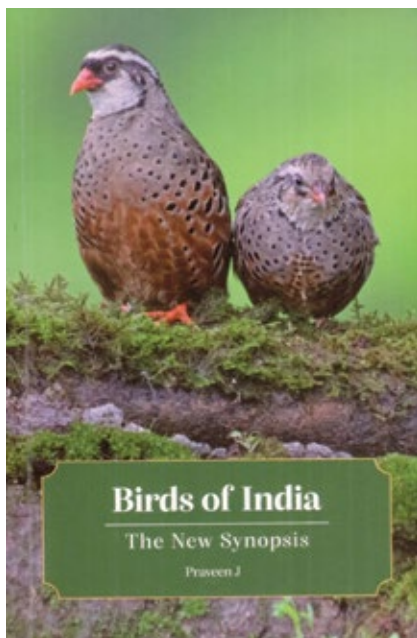
The editors also show how technology and accessibility have transformed birdwatching. The digital age – through platforms like *kolkatabirds.com*, *xeno-canto.org*, and *eBird* – has turned birding into a collaborative pursuit, connecting experts and amateurs. Notes in *Indian Birds* and *Sanctuary Asia* further amplify this network, turning once-isolated efforts into collective discovery.

However, the authors are candid about the challenges that remain: habitat loss, inadequate surveys, and the risk of species 're-disappearing' even after rediscovery. The case of Jerdon's courser underscores the need to move beyond the excitement of finding a species toward long-term conservation and ecological understanding.

Throughout its chapters, the book weaves meticulous field data with vivid first-hand narratives – from late-night expeditions for Mrs Hume's pheasant to arduous journeys through the Nicobar Islands and explorations of the Naga Hills and Manipur borderlands. These accounts show how improved access and growing local expertise have made once 'mythical' birds visible again, though still elusive.

In essence, the book is both a tribute to the past of Indian ornithology and a manifesto for its future. It celebrates those who refuse to accept that a bird is lost simply because it has gone unseen for decades. With equal attention to historical scholarship, modern field methods, and a spirit of adventure, it captures a changing India – where rediscovery is no longer the domain of a few colonial scientists, but the shared passion of a nationwide community. 📖





### **Birds of India – The New Synopsis**

by Praveen J.

Published by: Nature Conservation Foundation, 2025

Size: 21.5 x 14 cm

Pages: 468 + xlvii

Price: ₹ 499/-

Paperback

Reviewed by: **Harkirat Singh Sangha**

Praveen Jayadevan is passionate about birds. I first met him in 2013 on a pelagic birding trip and immediately noticed his infectious enthusiasm. Trained as a computer engineer, he later shifted to a full-time career in ornithology to pursue his passion. He joined the Nature Conservation Foundation, Mysuru, with the objective of compiling an updated checklist of the birds of India – a task long overdue. Over the last two decades, ornithology in India has seen a surge of activity, greatly expanding our knowledge of bird distribution and variation. The gap in the literature has now been filled by the author.

The first comprehensive taxonomic list for the Indian subcontinent, *A SYNOPSIS OF BIRDS OF INDIA AND PAKISTAN, TOGETHER WITH THOSE OF NEPAL, BHUTAN AND SRI LANKA*, was published in 1961 by S. Dillon Ripley and later updated by him in 1982. Forty-three years on, we are far removed from Ripley's work. *BIRDS OF*

*INDIA – THE NEW SYNOPSIS* by Praveen J. is much more than an up-to-date checklist. It attempts to summarise all advances in our knowledge of the distribution and nomenclature of Indian birds. Since Ripley's time, taxonomy has undergone considerable changes, including revisions in sequence and in the names given to India's geographical regions.

We have come a long way since the very first list published by T.C. Jerdon in his seminal *BIRDS OF INDIA* (1862). Allan Octavian Hume – political reformer, ornithologist, civil servant, botanist, and key founder of the Indian National Congress – published a checklist in his journal *Stray Feathers*. But it was E.C. Stuart Baker's eight-volume *FAUNA OF BRITISH INDIA: BIRDS* (1922–1930), which introduced the concept of subspecies, that set the tone for modern trinomial taxonomy. After Independence, Sálím Ali and S. Dillon Ripley published the monumental *HANDBOOK OF THE BIRDS OF INDIA AND PAKISTAN* in 10 volumes. This was followed by illustrated field guides by Richard Grimmett, Carol Inskipp, and Tim Inskipp (1998) and Krys Kazmierczak (2000). Pamela C. Rasmussen and John C. Anderton then produced the more comprehensive *BIRDS OF SOUTH ASIA – THE RIPLEY GUIDE* (2005; 2nd edition, 2012). More recently, a national checklist was published in *Indian BIRDS* (2016) and has been updated annually. At present, 1,375 bird species are definitively documented from India.

After three years of dedicated effort, *The New Synopsis* is now in our hands. What does it contain? It is a comprehensive, updated taxonomy of all 1,998 bird species and subspecies in India, covering 501 genera and 115 families reliably recorded within India's political boundaries up to December 31, 2024. The book includes acknowledgements, a list of acronyms, an introduction, notes on subspecies variation in Indian birds, family overviews, and an index. The 29-page introduction covers the history of Indian bird taxonomy, the updating of Ripley's synopsis, organisation of the book, taxonomy, sources, and future directions. The bulk of the work then examines each species on the Indian list, providing common and scientific names, alternate names, original descriptions (with author and date), taxonomy, distribution, specimens (and their locations), IUCN/WLPA/CMS status, and additional notes.

At last, we have a work of satisfactory scope – valuable not only to taxonomists but also to the growing community of birders, both amateur and professional. To the benefit of those who use eBird, the author has based his checklist on the widely adopted eBird/Clements taxonomy, including its sequence and nomenclature. For cross-reference, serial numbers from THE HANDBOOK are also given. Another key feature is the listing of institutions where Indian specimens are housed. Many new vagrants and rare species recorded in India are included for the first time.

As with any work of this scale, some errors are inevitable. I noticed a few slips while leafing through the book. For instance, the University of Michigan Museum of Zoology is mistakenly located in India rather than the USA (p. xiii). In some cases, distributions are given too casually. The red spurfowl (p. 17), confined to a small area in southeast Rajasthan, is listed as occurring across ‘Rajasthan, adjoining E Madhya Pradesh, and NE Gujarat’. Similarly, the painted spurfowl’s (p. 17) distribution is overstated as ‘west till about the driest parts of the Thar Desert’, though it is limited to the mesic eastern parts of Rajasthan.

There are also some technical errors in distributional assignments. The black-bellied sandgrouse (p. 37), which occurs only in western Rajasthan, is listed for ‘E Rajasthan’. Conversely, the grey-bellied cuckoo (p. 43), which occurs in eastern Rajasthan, is listed for ‘W Rajasthan’. The Eastern barn owl (p. 136) occurs locally in the Thar Desert, but is excluded from it here. And what is the basis for including the bar-winged flycatcher-shrike (p. 199) in Rajasthan?

At times, the distribution appears overly liberal. On the basis of single or rare stray records, the Asian houbara (p. 37) is listed for Kashmir and Kerala, and the cream-coloured courser (p. 85) is given as occurring in ‘S Kashmir’ in winter – both odd inclusions.

The reliance on eBird data is evident, sometimes at the cost of published literature. For example, despite many published records of the Wallcreeper (*Indian BIRDS*) from Agra, Fatehpur Sikri, Bharatpur, Etawah, and Bikaner, its winter range is given only ‘till Delhi’. Similarly, the Asian stubtail (p. 274) is listed as a ‘vagrant’ in India, even though Karimganj (Assam) is a regular site.

One last quibble: the book assumes readers are already well-versed in Indian geography. The inclusion of a map showing political and geographical regions would have made it more user-friendly.

These minor criticisms, however, do not detract from the importance of this work. At a time when websites and mobile apps threaten to make books redundant, this volume reminds us that for authoritative scientific reference, books remain indispensable. Ornithologists, zoologists, students, teachers, researchers, birdwatchers, and wildlife officials – all who seek accurate information on Indian birds – should have a copy.

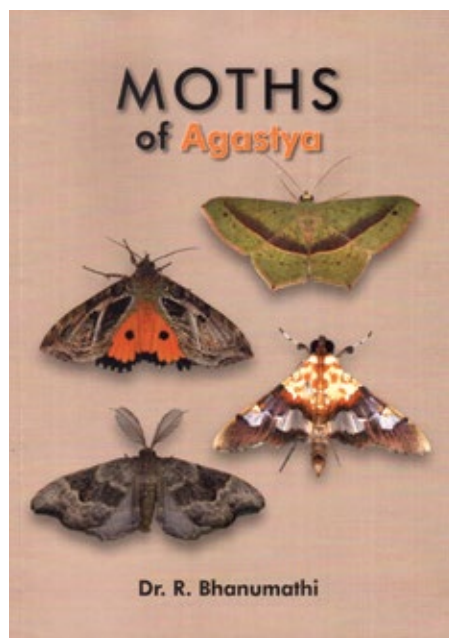
In short, this is a highly useful volume and an indispensable companion for anyone seriously interested in the taxonomy of Indian birds. Interestingly, the text is licensed under Creative Commons 4.0, making it freely usable with attribution. The paperback is available directly from the Nature Conservation Foundation (NCF), the book’s publisher. 📖

We are grateful to

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### **Moths of Agastya**

by R. Bhanumathi

Published by: Agastya International Foundation, 2025

Size: 21 × 15 cm

Pages: xvi + 216

Price: Not mentioned

Paperback

Reviewed by: **Kumaran Sathasivam**

All living things captivate the eager naturalist, and moths are no exception. Moths are a special group in that they are among the invertebrates that are large enough to catch one's attention even at a casual glance. Anyone who has the impression that moths are drab or dull will be disabused on flipping through this field guide.

Those who indulge in moth watching seriously, the term 'mothers' springs to the mind, have their own rituals. They wait for 'moon-dark' nights (around the new moon) and set up white screens in the open. They illuminate the screens with suitable lamps, mercury vapour and UV lamps being most effective. Then they maintain a nocturnal vigil, waiting for the screens to be peppered with waves of arriving moths. Author Dr R. Bhanumathi did this each month for four years at a place called Agastya. Not to be confused with a mountain peak in southern India, this Agastya is a 172-acre campus in Chittoor District, Andhra Pradesh.

A traditional entomologist or taxonomist would have collected specimens and subjected them to microscopic examination to identify the moths. In this day and age, is there any room for taking life just to satisfy scientific curiosity or to name an insect? There is growing recognition of the intrinsic value of all life forms. And there have been vast advances in photography in the last two or three decades. But moth study is lagging yet. One hopes that the day is not far away when new photographic technology is developed that permits even internal taxonomical characteristics to be studied without detriment to the moths.

However, in the pre-transition period that we are in, Bhanumathi is a pioneer. She took up the challenge of identifying the Agastya moths from photographs alone. Admirably, she did not capture or 'sacrifice' any of her subjects. She distinguished 653 forms. Of these, she identified 600 to the species level. This is no mean achievement.

The numbers might suggest to someone accustomed to birdwatching lists that Agastya is endowed with a particularly rich moth fauna. Is this so? The author mentions that some 15,000 moth species have been recorded in India, which provides a reference. But one needs the context of studies conducted in similar-sized campuses to make an assertion. Whatever one may conclude about Agastya as a moth habitat, the value of this book lies in its usefulness in a vast area outside this campus, which is after all a tiny speck on India's map.

The number of moth families Bhanumathi recorded in Agastya, and covered in the book, is 43. This number represents more than 40 percent of the moth families found in India. Thus, Bhanumathi's book will be useful to any naturalist seeking to identify moths to family level, even far from Agastya.

MOTHS OF AGASTYA is a sumptuous photographic documentation of Bhanumathi's findings, presented taxonomically. It is a visual treat to gaze at the moths, and the book has been tastefully produced. Incidentally, it has an unexpectedly solid feel in the hand.

You might conceivably use this pictorial guide in the field, when you are looking at live moths at a screen, but you would find it far more convenient to refer to it later when studying photographs you have taken.

With MOTHS OF AGASTYA in hand, you will be equipped to identify the higher taxa. Bhanumathi has provided

brief descriptions of the moth superfamilies, families and subfamilies. She has provided details such as size, markings, shape, etymology and habits though not all of these for each taxon.

The way moths fold and hold their wings when at rest creates a distinctive outline in many instances. In a preliminary step, the silhouettes of moths may be used to identify the families they belong to. Silhouettes, referred to here as 'resting postures', of moth families found at Agastya are presented. These are categorised as arrowhead shaped, bell shaped, triangular and so forth. However, the use of silhouettes to identify to family level is not an infallible method. In instances, you must seek further details.

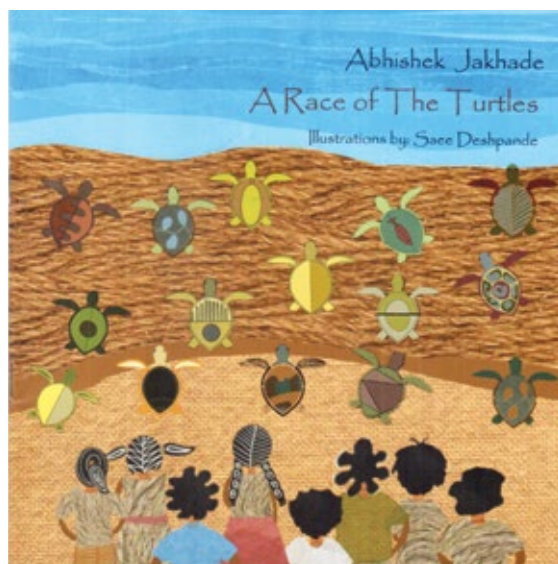
The book also lists for each family the wingspan range, antenna type and proboscis presence/absence and development. Presumably, each family has a unique combination of these features. Look for the proboscis!

What about identifying genera and species? Using this book, enthusiasts can even try identifying insects beyond the family level. The book includes photographs of all species recorded at Agastya by the author, with multiple views provided in some cases. The question is, are you sure you want to identify every member of this multitudinous group to genus or species level?

I selected moth photographs from my collection and attempted to identify the families using the book. It was a learning experience. If you do not have moth photographs, you could practise using the book with the photos of unidentified species it contains.

To sum up, you will be able to identify many Indian moths using this book. It will sharpen your observation, encouraging you to look closer at moths, revealing hidden diversity in familiar surroundings.

I recommend this book enthusiastically to anyone with an interest in natural history. 🐛



### **A Race of the Turtles**

by: Abhishek Jakhade

Published by: Padmagandha Prakashan

Size: 20 x 20 cm

Pages: 36

Price: ₹ 180/-

Paperback (available in English and Marathi)

Reviewed by: **Priyadarshini Supekar**

This beautifully crafted story is a heartfelt tribute to both nature and the quiet heroes who dedicate their lives to protecting it. What begins as a simple race unfolds into a deeply moving journey, illuminating the tireless work of turtle conservators who safeguard delicate nests and guide vulnerable hatchlings toward the sea.

Ideal for young readers, the book seamlessly blends nostalgia with environmental awareness, gently reminding us that even the smallest acts – shielding a nest, clearing a path – can leave a lasting impact. Its serene beach setting and lyrical storytelling allow you to almost see the hatchlings, feel the warm sand, and hear the ocean calling them home.

Reading it stirred vivid memories of watching olive ridley hatchlings inch toward the waves, their tiny flippers pushing forward with determination, the sea welcoming them like an old friend. One feels like cheering on these 'little explorers' scurrying toward the shore; each tiny turtle, brave yet wobbling, fills the observer with joy, hope, and curiosity. That emotional resonance turns this book into more than a story – it becomes an experience.

A must-read for young nature lovers, educators, and anyone who believes in the quiet magic of conservation. 🐢



## Indian Cormorants Hunt Bullfrogs in Junagadh's Wetlands



In the tranquil wetlands near Lal Dhori – nestled at the foothills of Girnar in the Bhavnath area of Junagadh, Gujarat – I watched Indian Cormorants *Phalacrocorax fuscicollis* display remarkable hunting behaviour as they went after bullfrogs.

These sleek waterbirds, known for their expert diving and fishing skills, were fascinating to observe. Instead of chasing fish, as they usually do, they were targeting bullfrogs – large, muscular amphibians better known for catching insects than for being caught themselves.

Despite their size and powerful legs, the bullfrogs were no match for the cormorants' stealth and agility. I watched the birds dive beneath the surface, propelling themselves with strong kicks of their webbed feet. Their slender, hooked bills were perfectly shaped to seize and swallow prey whole, making them impressively efficient hunters.

What struck me most was how this behaviour revealed their adaptability. Although cormorants mainly feed on fish, they are opportunistic feeders, ready to take advantage of whatever prey is most available – whether it's fish, crustaceans, or even amphibians like bullfrogs. This

flexibility helps them thrive in diverse habitats such as the rich wetlands around Lal Dhori.

The cormorants I observed chased Indian bullfrogs *Hoplobatrachus tigerinus*, diving and gliding swiftly through the water to ambush them. Once a frog was within reach, the bird lunged, grabbed it with its hooked beak, and surfaced to swallow it headfirst – an efficient way to keep the frog's limbs from catching in its throat.

I noticed they caught both young and adult bullfrogs. The smaller, slower juveniles were easier prey, but larger adults – sometimes up to 17 cm long – were taken by the bigger species such as the Indian cormorant and the great cormorant. Watching this, I was reminded of how finely tuned these birds are to life in freshwater habitats.

By observing their feeding patterns, I have come to appreciate how much such moments reveal – not just about bird behaviour, but also about wetland ecology and the intricate relationships that sustain our natural world.

– Mihir Ranva

Junagadh, Gujarat Member O-73260

# William Mansfield, Lord Sandhurst

3rd President of BNHS (1895–1900)



**William Mansfield**, the 1st Baron Sandhurst and the 1st Viscount Sandhurst (1855–1921), also known as Lord William Barron Sandhurst, GCIE, served as the Governor of the Bombay Presidency from 1895 to 1900. He was the 3rd President of the Bombay Natural History Society. Following his term as Governor, he was appointed Knight Grand Commander of the Order of the Star of India (GCSI) in 1900.

Lord Sandhurst's tenure as Governor was marked by the outbreak of the bubonic plague in Bombay and its surrounding areas during the summer of 1896. By March 1897, more than 20,000 lives had been lost, and residents were rapidly fleeing the city limits.

In the decades following the Sepoy Mutiny of 1857 and other uprisings, British officials in India were often hesitant to impose strict public health measures – a tendency that persisted into the 1890s. In a letter to Lord Hamilton, the Secretary of State for India, Lord Sandhurst attributed the unreadiness of the inhabitants and their distrust of sanitary interventions. In February 1897, he enacted the Epidemic Diseases Act, empowering authorities to combat the plague. This legislation ultimately led to the development of sanitation, drainage, and sewerage systems in Bombay – many aspects of which remain operational in current times.

Uniquely, and in recognition of the measures that he implemented and stands to this day in Bombay (=Mumbai), is the construction of the Freemasons Hall, constructed

in June 1897 and still in use today. Located opposite Sterling Cinema, this building temporarily housed the city's municipal offices while the Municipal Corporation Building was under construction. This development was made possible through the unprecedented appointment of the Governor of Bombay as the District Grand Master for both the English and Scottish Masonic Constitutions. During his tenure, the Bombay City Improvement Trust was also established in 1898, leading to the construction of the city's now-iconic chawls – tenement-style housing that improved sanitation and living conditions in increasingly congested urban areas. Lord Sandhurst himself laid the foundation stone of Bombay's first chawl.

The plague pandemic, which began in Bombay in 1896, spread across British India and persisted for three decades, claiming an estimated ten million lives. From Bombay, it spread to Poona, Karachi, and reached Calcutta the following year. The authorities responded with quarantines, isolation camps, travel restrictions, and demolition of infected buildings – measures that provoked significant public resistance. During Lord Sandhurst's tenure, the Pasteur Institute identified the role of the rat flea in transmitting the disease. Soon after, Waldemar Mordecai Haffkine developed the first plague vaccine, famously testing it on himself before saving millions of lives.

The popularly recognised city landmark is the area known as Sandhurst Road, named in recognition of his efforts to combat the plague. The area was a flagship development of the Bombay Improvement Trust, extending from the eastern docks to the western seaboard. Many migrant and local workers were housed in the newly built chawls along this stretch.

A lesser-known fact is that Sandhurst Road Station on Mumbai's Harbour Line was originally known as Mazgaon Station. With the electrification of the suburban railway network in the 1920s, the old Mazgaon station was relocated southward and renamed after Lord Sandhurst. Consequently, the suburbs of Mazgaon and Umerkhandi came to be known collectively as Sandhurst Road.

In August 1899, the Governor House in Parel – formerly a Jesuit chapel – was handed over by Lord Sandhurst to Dr Waldemar Haffkine, a student of Louis Pasteur, to establish the Plague Research Laboratory. It was renamed the Bombay Bacteriology Laboratory in 1906, and later, in 1925, became the Haffkine Institute, a name it proudly bears today.

*Abridged from various public sources.*

– Dr Bharat Bhushan  
Honorary Secretary, BNHS



# Underwater Realms

Text and Photographs: **Dhrishit Khandhar**



Known as the guardian of coral reefs, the Hawksbill Turtle feeds on sponges and prevents their overgrowth coral to thrive

The ocean has always felt like home to me – a sanctuary of calm and wonder. Exploring the vibrant underwater world had long been a dream, one I finally fulfilled during a mesmerising trip to the Andaman and Nicobar Islands. Diving was not just an adrenaline rush – it offered a humbling, eye-opening glimpse into the delicate balance of marine ecosystems and deepened my understanding of why they desperately need our protection. From darting fish to graceful turtles, each encounter left an indelible mark on my memory.

Located in the Bay of Bengal, off India's eastern coast, the Andaman and Nicobar archipelago comprises a staggering 349 islands, most of them untouched or inhabited by indigenous tribes. Spanning a total coastline of 1,962 kilometres and covering 8,249 square kilometres, the islands are a treasure trove of biodiversity both above and below the surface.

Our journey began in the bustling capital of Port Blair, from where we took a ferry to the island of Swaraj Dweep (formerly Havelock Island). Our first dive site – Kalapatthar Beach – lay on the southeastern shore. This sandy stretch, punctuated by scattered basalt rocks, revealed an underwater



Once a capsized cargo ship, the Inchkeith wreck is now a reef, covered in vibrant artificial corals and sponges where small fish hide among its chambers

landscape teeming with life. Verdant mountains framed the coast, offering a picturesque scene both on land and beneath the waves.

Descending into the ocean, I was struck by the stillness found at greater depths. With each gentle kick, I drifted past shoals of moorish idols, sergeant major fish, and parrotfish, who swirled around me in bursts of colour. Near the end of the dive, my buddy pointed out an octopus – nearly invisible at first thanks to its flawless camouflage. Mimicking the texture of nearby rocks, it had hardened its gelatinous form, only to vanish in a puff of ink when it sensed our presence. This fleeting encounter felt like witnessing nature's own Houdini at work.

Another dive brought us to the Inchkeith Wreck, a hauntingly beautiful site near Havelock. The ship's stern lay just five metres below the surface, while the bow reached down to 30 metres. Originally a cargo vessel delivering coal and timber from Kolkata to Mumbai, it met its end in 1955 when it struck shallow rocks in stormy weather. Now, decades later, it has transformed into an oasis of life.

Almost immediately, a majestic hawksbill turtle glided past, its intricately patterned shell blending seamlessly into the corals and rocks. I watched as it fed on sea sponges growing along the ship's rusted exterior, its hooked beak working with surgical precision. Critically Endangered and the only surviving species in its genus, the hawksbill's presence made the dive truly special.

We explored the ship's inner chambers, where shafts of light filtered through the rusting skeleton. Coral colonies, squid, and triggerfish had claimed the wreck, while juvenile fish sought refuge inside the hull. Nearby, vibrant coral polyps swayed gently, hosting clownfish and cleaner fish that played their vital roles in a delicate web of symbiosis. A half-eaten sergeant major fish lay nearby – a stark reminder of the raw, unforgiving side of life underwater.

One of the most magical moments came during a dive at South Button Island, a remote national park we reached after a two-hour ferry ride. As we descended into the blue, I found myself face-to-face with two manta rays – oceanic giants that moved with astonishing





Protected by a special mucous layer, clownfish thrive among anemone stings – and in return, they clean and protect their host



Growing over four feet long, moray eels are nocturnal hunters that feed on invasive lionfish, swallowing them head-first to avoid the venom

grace. Their vast wings seemed to propel them through water like birds in slow motion. At one point, a manta soared directly above me, casting a massive shadow as I caught sight of its feathered gills undulating in the current – an unforgettable, almost surreal experience.

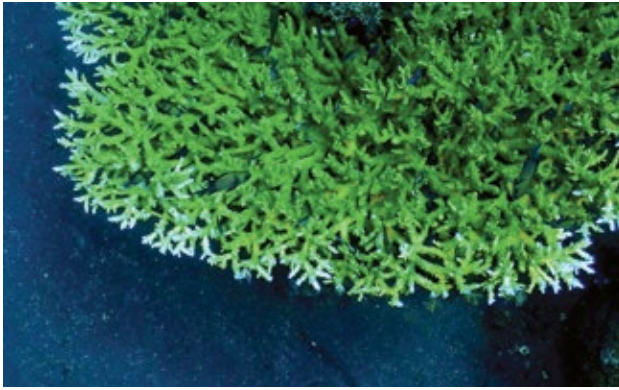
Later, we encountered a yellow-edged moray eel, its ribbon-like body tucked between speckled rocks. Stealthy and cautious, it peered at me warily

as I tried to inch closer for a photograph, my heart racing at the thought of getting too close.

Above the surface, Havelock's mangrove walk offered a striking contrast. Meandering along wooden boardwalks, I passed tangled roots rising from brackish water, supporting an ecosystem as vital as it is vulnerable. Spanning 14 square kilometres, Havelock's mangroves serve as natural barriers against erosion and



A perfect symbiotic relationship – mangroves protect reefs from pollution and sediment, while corals reduce the wave energy, shielding mangroves from damage



Stony corals first appeared in the Middle Triassic (201 million years ago) and have been the main reef-building organisms ever since



Often misunderstood as dangerous due to their size and resemblance to stingrays, manta rays are gentle and gain nutrition through filter feeding



A lionfish can expand its stomach more than 30 times in volume when consuming a large meal and can survive without food for over 12 weeks

sanctuaries for countless species. Yet they face mounting threats from deforestation, coastal development, and climate change. Over the past few decades, approximately 2.5 square kilometres have been lost.

Conservation efforts are underway, from reforestation programmes to community awareness campaigns. Underwater, however, the stark reality of environmental degradation is hard to ignore. Bleached corals and floating waste offer painful evidence of a warming world. Corals, sensitive to temperature shifts, absorb excess UV radiation, generating toxic oxygen compounds that damage themselves and their neighbours.

Yet hope remains. I witnessed a coral undergoing micro-fragmentation, a technique where damaged corals are broken into small pieces and mounted on frames to encourage rapid growth. This glimpse of restoration reminded me that change is possible. Across the

globe, marine biologists, conservationists, and local communities are working tirelessly to tip the balance back in nature's favour.

As I surfaced from my final dive, David Attenborough's quote echoed in my mind:

*"No one will protect what they don't care about, and no one will care about what they have never experienced."* 🐠



**Dhrishit Khandhar** is an avid wildlife photography enthusiast. He recently authored a literature review on coral restoration for the *International Journal of Health Sciences and Research*.



# A Nest of Surprises: The Purple-rumped Sunbird of Goalpara

Text: **Rupjyoti Rabha and Sachin Ranade**

Photographs: **Rupjyoti Rabha**



Purple-rumped Sunbird (male)

It was a bright morning in May 2024, when the first author saw a pair of purple-rumped sunbird *Leptocoma zeylonica* in the backyard of his home in Goalpara, a bustling town in western Assam. The tiny architect was at work, weaving her delicate hanging nest.

We have always admired these little gems of the tropics. Barely the size of a thumb, the female sunbird is a master builder – meticulous, inventive, and fiercely independent. She builds her pear-shaped home entirely by herself, using spider webs, fine fibres, and bits of plant fluff, crafting a nest that sways gently from a low branch or a convenient wire. The male, with his iridescent plumage, is more of a supervisor – often flitting nearby but rarely lifting a feather to help.

At first, everything about this nest seemed straight out of an ornithology textbook – the silk-bound frame, the hanging design, the neat oval entrance. But then something unexpected caught our eye, plastic: Among the natural fibres, shiny strips of discarded plastic had been woven into the walls of the nest, glinting faintly in the sunlight. We could not help but smile at the irony. Perhaps, we joked,

the bird thought the nest would look too artificial without a bit of plastic in it!

Still, despite our amusement, it was a subtle reminder of how deeply human materials have infiltrated even the most delicate corners of nature. Yet, the finished nest was a masterpiece – beautifully camouflaged, blending into its surroundings so well that any passing predator would have easily overlooked it.

Days passed, and the nest remained untouched. No eggs, no signs of incubation. The pair visited occasionally but never seemed to settle in. We waited, wondering what had gone wrong. Then, about twenty days later, a surprise unfolded. A juvenile sunbird appeared in the same garden, fluttering about confidently. Curious, we began a careful search – and just five feet away from the first nest, we found another, newly built structure. It seemed that the pair had built a second nest and used that one instead.

Was the first nest a ‘dummy nest’ – a decoy to distract predators or a failed attempt? Similar behaviours have been documented in wrens and a



Purple-rumped Sunbird female with nest material

few other bird species, where the first structure serves as a false lead. Could the purple-rumped sunbird be employing a similar strategy? The possibility was fascinating and certainly worth further observation.

The observation also carried scientific interest for us. Goalpara lies slightly north of the usual distribution range of the purple-rumped sunbird, suggesting a subtle range expansion. The bird’s use of anthropogenic materials – in this case, plastic – echoes a growing global pattern. A recent study documented over 6,000 bird species worldwide incorporating human-made materials into their nests, a trend both adaptive and concerning.

Here, in one small garden in Assam, that story was being rewritten – a testament to the resilience and adaptability of birds in a changing world. We could not help but admire the ingenuity of the birds. Whether building a perfect home or creating a clever decoy, these little creatures reminded us that nature is endlessly inventive – and that even in a world reshaped by humans, life finds new ways to adapt, survive, and thrive. 



**Rupjyoti Rabha** is an ardent bird watcher since his childhood. He has been documenting birds, primates and effect of highway-traffic on the wildlife in Goalpara district, Assam.



**Sachin Ranade** is Assistant Director at the BNHS, working on vulture conservation for last 25 years. He has keen interest in the various taxa and their biology.



# MUDSKIPPER

## – the fish that walk

Text and Photographs: **Subir Ghosh**

**I**n the Sundarbans Mangrove, the apex tourist attraction is, without a doubt, the apex predator – the tiger. Next in the pecking order come other mammals: leopards, fishing and jungle cats, short-clawed otters, and deer. Reptiles such as saltwater crocodiles, monitor lizards, and snakes, along with a dazzling variety of birds and lush vegetation, also command attention – each according to one's interests.

Yet, more often than not, an intriguing species escapes notice – the mudskipper. During a visit to the Sundarbans, I was determined to take a closer look at these lesser known.

Mudskippers are barely visible from the decks of regular tourist vessels, as visitors are not permitted to set foot on the riverbeds. Their small size (adults grow to about 25–30 cm) and muddy coloration blend seamlessly with the flats that emerge during low tide. I had expressed my interest in photographing these fascinating fish to Ayan Mondal, our government-registered forest guide, and Himadri Mondal, a well-known wildlife photographer and proud owner of our floating home, The Mangrove Mist. They explained that mudskippers thrive in mudflats made of silty soil – well-drained yet moist enough for burrow-digging. Heavier, stickier silt, like the kind we were passing through, does not suit them.

Eventually, we anchored near a shore – lured by a WhatsApp alert of a tiger sighting – and found ourselves in an area teeming with mudskippers, both juvenile and adult, all busily engaged in their daily routines.

Colour display during courtship or fight





Territorial dispute



Staying near the mudhole burrow

### What Are Mudskippers?

Mudskippers belong to a unique group of amphibious fish in the subfamily Oxudercinae of the goby family. This group comprises 23 extant species found across the Indo-Pacific – from Africa to Polynesia and Australia – in swamps, estuaries, and tidal mudflats.

These fish-like creatures, with their elongated bodies, spend more than half of their lives on land, engaging in everything from feeding to mating out of water. Their close-set, bulbous eyes – perched like periscopes atop their heads – can blink to clean and moisten themselves.

One of their most remarkable traits is their ability to climb, walk, and even jump on land. This terrestrial movement is made possible by their forward-set pectoral fins, which act

almost like limbs, allowing them to crawl. During low tide, they emerge to forage on crustaceans, small crabs, insects, snails, and sometimes even other mudskippers.

### Masters of Amphibious Adaptation

Mudskippers have evolved extraordinary adaptations that allow them to survive out of water for long periods. They breathe using water bubbles trapped in their enlarged gill chambers and can also absorb oxygen through the linings of their mouth and throat. In addition, like amphibians, they engage in cutaneous respiration – breathing through their skin, which is rich in capillaries. These adaptations require a consistently moist environment, explaining their preference for estuaries and mudflats.

As I sat on the boat, trying to photograph them with a telephoto lens, I was astonished to see their otherwise brownish-green bodies

shimmer with bright blue spots – an enhancement developed during the mating season to attract females. I soon became engrossed in a territorial





A Collared Kingfisher sent the mudskippers diving tail-first into their burrows



Eurasian Curlew – one of the many bird species seen in the Sundarbans

duel between two adult males, their bodies dazzling with speckled blue markings and fully splayed dorsal fins. They postured, yawned wide to intimidate, bit fiercely, and even drew blood. At times, they locked jaws and pushed each other like miniature sumo wrestlers battling for dominance.

In what initially seemed like another scuffle, I realized that the mudskipper was not fighting a rival but courting a potential mate – his dominance display transformed into an elaborate dance of attraction.

Suddenly, a collared kingfisher – a known predator of mudskippers – appeared nearby. In an instant, the mudskippers vanished into their burrows, retreating tail-first while keeping only their eyes and mouths visible above the mud. These burrows are vital to their survival. During high tide, they retreat inside, surviving in oxygen-poor water by maintaining an air pocket. Males also use these burrows to court females, enticing them inside to lay hundreds of eggs. Once the

female departs, the male stays behind, guarding the brood from predators.

As we prepared to leave, I found myself reflecting on the evolutionary significance of these extraordinary creatures. Mudskippers may hold clues to one of the most fundamental questions in evolutionary biology: how vertebrates made the monumental leap from water to land. Watching them, I felt as though I had witnessed a living chapter of life's great evolutionary story – right there in the tidal mud of the Sundarbans. 



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# ISLANDS OF HOPE

Text and Photographs: **Khushboo Rani**

**T**he Ganga flows gently in summer, its waters calm and low, revealing wide stretches of golden sandbars that gleam under the rising sun. These sandbars – ephemeral islands born from the river’s retreat – emerge quietly every March within the Vikramshila Gangetic Dolphin Sanctuary (VGDS) in Bihar. To most people, they appear lifeless and barren, just stretches of sand lost between channels of water. But to me and my team, these are extraordinary theatres of life – seasonal homes where birds return year after year to lay eggs, raise their chicks, and defy the odds on open, vulnerable terrain.

My relationship with this river and its birds began in 2021, when I first stepped into VGDS as part of

the BNHS Wetlands and Flyways Programme. A year later, in 2022, we began systematic monitoring of sandbar-nesting birds across a 65-kilometre stretch between Sultanganj and Kahalgaon. I still remember the quiet awe of spotting our very first nest – a simple scrape in the sand holding three speckled eggs. A little tern hovered above, torn between guarding its nest and trying to distract us from it. In that moment, I realised that this was not just another field site. This was a living system that demanded long-term attention. We needed to return season after season to truly understand its rhythms.

From our observations, we learned that the breeding season begins in March, as pairs arrive and

Golden sandbars emerge as the Ganga recedes in summer, forming the vulnerable, seasonal homes where sandbar-nesting birds return year after year to breed in the VGDS







Little Pratincole at an open nest on the sandbar

start preparing their nests. By early April, when the river has further receded, the sandbars open up completely, and the real drama of survival begins. For the next three to four months – until the monsoon arrives in June – these temporary islands become cradles of life.

Our field days start before sunrise. Bihar lies in India's eastern stretch, so dawn breaks early – sometimes as early as 4:00 a.m. After a quick

team briefing and a strong cup of tea, we load our equipment onto a small boat at Bhagalpur and set out onto the calm waters. Depending on the survey plan, our boarding points change daily. Sometimes we travel long stretches by vehicle, where boatmen await us at pre-decided spots to begin the day's surveys.

Every move we make during this time is careful and deliberate. We try to minimise disturbance to the breeding pairs, who are often just metres away. To prevent overheating of exposed eggs or forcing adult birds into unnecessary flight, we restrict our surveys to the cool hours between 4:30 and 9:00 a.m.

The river at dawn feels sacred – wrapped in stillness, with only birdsong and the gentle hum of the motor breaking the silence. As we drift past sleepy villages and morning smoke curling from thatched roofs, the landscape feels timeless. Once we reach a sandbar, the work begins.

Our team walks the length of each sandbar, either following random observation routes or belt transects, depending on its size and terrain. We scan for nests, eggs, and chicks, marking each location with GPS and noting details – species, habitat type, human activity, and distance from



Bank Mynas and their nesting holes in a sandbank cliff along the river – an important breeding habitat for landbirds

water. Sometimes, we sit quietly at a distance, observing parental behaviour without intruding.

That year, we recorded 140 nests belonging to five species. We soon realised that we needed more hands to survey effectively – the birds were nesting across 13 of the 22 islands we had covered.

In 2023, we trained 54 Forest Department frontline staff, Dolphin Mitras, and Wetland Mitras to join our efforts. We documented 167 nests of seven species – and, most excitingly, the Indian skimmer bred here for the first time in 17 years. By then, we knew the river so intimately that we could almost predict where each species would nest.

In 2024, we expanded further, involving interns from Guru Ghasidas Vidyalyaya, Bilaspur, and a master's student from Patna University. Together, we completed full surveys of all 13 sandbars, recording 866 nests of seven species. The small pratincole dominated in 2022, little terns in 2023, and again small pratincoles in 2024.

The work is beautiful, but it is also demanding. The sun can be relentless, and walking for kilometres across burning sand is exhausting. There are no trees, no shade, and often not even a breeze. Some days we return with notebooks

full of data; other days, the sandbars seem empty, silent, holding their secrets close.

There are heartbreaks too. Sometimes buffalo herds trample entire colonies. Other times, an overnight water release from upstream dams floods the islands, washing away nests we had mapped just the previous day. Over time, we began analysing how factors like elevation, location, and disturbance levels affected nesting success.

The threats, we found, are many. Sandbars close to towns are frequently used by livestock. Sand mining operations destroy nesting grounds. Fishing, boating, sewage outfalls, cremation rituals – all disturb these fragile colonies. To address this, we began mapping threats and human pressures using GIS, hoping to guide conservation interventions. We soon realised that data alone was not enough. Conservation on this river would depend on the trust and participation of its people.

In 2023, we began outreach efforts in the riverside villages – meeting boatmen, fishermen, panchayat leaders, and farmers. We showed them photographs of the birds, explained their importance, and discussed how local cooperation could protect them. Slowly, change began.



Residents and their water buffalo bathe and graze along an exposed sandbar – a frequent human-livestock disturbance affecting the fragile nesting colonies in the VGDS





Training and engagement efforts with residents and Forest Department officials for systematic monitoring of sandbar-nesting birds in VGDS

One morning in 2025, while we were near Bishanpur, a boatman named Amar came running up to me, beaming. “Didi”, he said proudly, “the Indian Skimmer has returned this year!” His excitement was genuine, and in that moment, I realised that the project was no longer just ours – it belonged to the community too. Today, several locals act as informal guardians, keeping livestock away from colonies and alerting us about disturbances or new nest sightings.

Working on the river has changed me profoundly – not just as a scientist, but as a person. As a woman leading a field team in a largely male-dominated space, I have faced my share of challenges. Our small boats have no washrooms, and on long survey days, I often went hours without drinking water or had to wait until we reached an island to find even a shred of privacy. But over time, I noticed a shift in how people perceived me. They began calling me “Didi” – a term not just of respect, but of belonging. I had become, in their eyes, someone who truly belonged to the river.

The Vikramshila Gangetic Dolphin Sanctuary was established in 1991 to protect the endangered Gangetic dolphin. But over time, we have learned

that its ecological richness extends far beyond a single species. This 65-kilometre stretch of the Ganga – its shape constantly changing with the flow – is a mosaic of deep channels, braided riverscapes, backwaters, and sandbars. It supports not only dolphins but also the Critically Endangered gharial, vulnerable smooth-coated otters, several species of freshwater turtles, and more than 200 species of birds.

Despite its protected status, the sandbars – so vital for nesting waterbirds – still lack focused conservation efforts. These islands exist for only a few months each year and provide rare open space for about a dozen sandbar-nesting species, including little terns, small pratincoles, Indian and great thick-knees, river lapwings, black-winged stilts, red-wattled lapwings, little ringed plovers, and greater painted-snipes. It was this fragile window of opportunity – the brief nesting season – that made VGDS the perfect site for our long-term study. And it rewarded us with the triumphant return of the Indian skimmer after seventeen long years.

While our main focus was on the mid-river sandbars, the riverbanks revealed another fascinating story. In 2022, during an early-season survey, I noticed something unusual on



A Black-winged Stilt chick fitted with a metal ring and leg flag, was seen along a shallow grassland edge



Little Terns observed at a sandbar nesting site. An adult carrying a prey item indicates active breeding and provisioning

an eroded mud wall – a group of Asian plain martins tending to their nests, visibly agitated. Soon we discovered why: bank mynas were invading the same nesting holes. The martins, which usually breed from October to February, had left behind burrows that the Mynas were now reusing during their own breeding season from March to June.

This overlap intrigued us. It was a moment that expanded our perspective. Beginning in 2024, we launched a dedicated survey of landbirds nesting along riverbanks, focusing on species like the bank myna, blue-tailed bee-eater, green bee-eater, and Asian plain martin. River bank surveys come with their own challenges – steep, crumbling slopes, and limited access points – but they have opened an exciting new chapter in our work. Understanding how these species interact, compete, and adapt in such a dynamic environment adds a deeper layer to our understanding of VGDS.

After three seasons of monitoring, we have built a strong dataset and identified 13 ‘safe nesting sandbars’ that could be protected during the breeding months. We have proposed seasonal protection measures to the Divisional Forest Officer, Bhagalpur – restricting sand

mining, reducing boat traffic, and involving local communities to keep cattle away between March and June.

All these birds need is a little space, a bit of peace, and a chance. As I stand on the riverbank today, watching the first light glint off the water, I feel an immense sense of gratitude – for the river, the birds, and the people who have become part of this journey. The Ganga has taught me patience and resilience. Each season brings both loss and renewal, but always, the birds return. And as long as they do, so will we. 🐦



**Khushboo Rani** is a Project Associate at the BNHS.

A passionate conservationist, she has extensive experience in bird migration and wetland ecosystem studies.



# Why Grasslands are ignored in India?

Text: **Kishor Rithe**

**G**rasslands are among the most widespread and yet most misunderstood ecosystems on our planet. These terrestrial landscapes dominated by herbaceous and shrub vegetation, shaped and sustained by fire, grazing herds, drought, light, wind velocity, and even extreme temperatures that prevent trees from taking hold. The term covers both the open grasslands with less than ten percent tree cover and the savannas with 10-15 percent, where a light scattering of woody plants grows among well-developed grasses.

These sweeping ecosystems occur on every continent except Antarctica and some larger islands, taking on many forms – from dense tropical bamboo stands to the treeless northern steppes, from the dry central plains to the windswept grasslands near the Arctic. Revenga and his colleagues in 1998 observed that a remarkable twenty-five of the world's 145 major watersheds are dominated by grasslands, each covering at least half of its area. Davis and co-researchers, in their 1994 survey of Centres of Plant Diversity, found that around 234 of these centers included grassland

Padharpaoni meadow in Tadoba-Andhari Tiger Reserve, Maharashtra





habitats, representing a vast storehouse of plant species and conservation potential. Stattersfield's team, writing in 1998, added that 23 of the world's Endemic Bird Areas depend primarily on grasslands. And Olson and Dinerstein, the same year, identified 35 terrestrial ecoregions as outstanding examples of these unique ecosystems.

Beyond their biodiversity, grasslands are immense carbon vaults. A study by the US Geological Survey in 1999 estimated that they hold roughly 34 percent of the planet's terrestrial carbon – nearly rivalling the 39 percent stored by forests and far surpassing that of agricultural systems. In this quiet way, the world's grasslands help to buffer the climate itself.

### Grasslands in India

In the Indian context, the story becomes even more intricate. Although India supports one of the largest livestock populations on Earth – 535 million animals, as recorded in 2019 – the country has never truly developed a coherent system for scientific pasture management. The subcontinent's ecological range is immense, from the humid plains of the western coast to the frozen deserts of Ladakh. As Mani noted in 1974, India sits at the crossroads of three major biogeographic realms: the Palaearctic, the Malayan, and the Ethiopian. This intersection

gives rise to extraordinary biodiversity, making India one of the twelve recognized mega-diversity centers of the world, as Koshoo observed in 1994.

A global assessment by Environmental Systems Research Institute (ESRI) – a provider of Geographic Information System (GIS) solutions in India – in 1993 and the Global Land Cover Data Repository (GLCDR) in 1998 estimated India's total grassland area at more than 534,000 square kilometres. Champion and Seth, writing in 1968, described Indian grasslands as 'pre-climax' communities – ecosystems that often exist as secondary stages in succession, yet remain stable for centuries under the combined influence of grazing and fire. They classified the Shola grasslands as subgroup II A, type C1, acknowledging their unique place in the country's ecological mosaic.

Fires in protected areas (PA) and grazing in unprotected lands have long been the primary forces maintaining these tropical grasslands. In 1973, Dabadghao and Shankarnarayan identified five principal grass cover types across India: the Sehima–*Dicanthium* type, the *Dicanthium*–*Cenchrus*–*Lasiurus* type, the *Phragmites*–*Saccharum*–*Imperata* type, the *Themeda*–*Arundinella* type, and the temperate–alpine cover of the Himalaya.

These ecosystems spread across Gujarat's Saurashtra, the Thar Desert of Rajasthan, the Deccan



A grassland in Kisli meadow in Kanha Tiger Reserve, Madhya Pradesh

KISHOR RITHE





Bengal Florican



Great Indian Bustard searching insects in grasslands in Pokharan

plateau of Maharashtra, the Malwa plateau of western Madhya Pradesh, the Terai belt of Uttar Pradesh, and the Brahmaputra Valley in Assam. Mountain grasslands also flourish in the Western Himalaya, the uplands of Sikkim, and the high slopes of the Western Ghats.

### Studies and Observations

While grassland ecology has been widely studied in the West, Indian research has traditionally focused on plant taxonomy rather than ecosystem dynamics. Few studies have explored the relationships among species, their habitats, and the environmental forces – frost, fire, and grazing – that shape them. High-altitude grasslands, despite their ecological significance, have remained largely unstudied.

The Bombay Natural History Society (BNHS) made early efforts to fill this gap. Between 1991 and 1997, BNHS carried out a study on the ecology of Indian plains grasslands, focusing particularly on endangered fauna. Supported by the Ministry of Environment and Forests (MoEF) and the U.S. Fish and Wildlife Service, the study revealed the fragility of these landscapes. Later, from 2000 to 2005, BNHS turned its attention to the Shola and alpine grasslands of southern India, examining their ecology and the mounting threats from human activity.

The findings were troubling. In the Nilgiris, for instance, once-pristine habitats were found to have suffered immense degradation. Much of this, the studies noted, resulted not from the indigenous Todas

but from later settlers such as the Badagas, who had inhabited the hills for about four centuries and were historically associated with extensive forest clearing. The introduction of exotic species – eucalyptus and acacia, brought from Australia for firewood and pulp – further altered the landscape, while the spread of tea plantations and hydroelectric projects submerged valleys and displaced native flora and fauna. By the mid-twentieth century, the Nilgiris had become one of the most ecologically transformed regions of India.

### Vanishing Birds and Changing Lands

BNHS researchers also documented the decline of the lesser florican, a charismatic grassland bird once widespread across Rajasthan, Gujarat, and Madhya Pradesh. Its breeding grassland habitats have been steadily lost to intensified agriculture, land encroachment, infestation of invasive species and shifting crop patterns. The florican was left with no option to adopt the traditional crops such as moong, and urad (occasionally jawar, bajara too) – plants that grow to modest heights of few feet – but farmers, now turning to cash crops, have left little space for these. Mechanized farming, heavy pesticide use, and the invasion of *Prosopis juliflora* have made matters worse.

Overgrazing in the remaining grasslands leads to trampling of florican nests, while the conversion of pastures into plantations and croplands deprives the species of safe breeding sites. To counter this, BNHS has been working to restore degraded habitats in

Rajasthan, raising funds to convert cropland back into grassland in both breeding and non-breeding zones. Based on BNHS recommendations, the Rajasthan Forest Department halted mining in key florican areas – notably in Ajmer and Bhilwara districts. By March 2023, a Conservation Reserve covering nearly 900 hectares had been established in Ajmer’s Arwar–Goyala–Kheeriya region, a rare victory for India’s grassland species.

Yet other dangers remain. The expansion of power lines across grassland habitats has proved lethal for the great Indian bustard and other low-flying birds, whose collisions with wires are now a leading cause of death.

### Policy Gaps and the Search for Solutions

Grasslands have evolved under cycles of grazing, drought, and fire—a balance that Anderson highlighted as early as 1982. Fire, when used judiciously, remains one of the most effective tools for maintaining grassland health, especially in PAs. But in India, where livestock graze freely across public and private lands, no clear grazing policy exists. Uncontrolled grazing, coupled with administrative neglect, has led to widespread degradation of grasslands.

Nearly half of India’s 329 million hectares of land is considered degraded. Massive afforestation programmes have been launched under social

forestry and wasteland development schemes, but many have failed because of overgrazing, unsuitability of lands for plantations and poor planning. Before Independence, forest grazing was tightly regulated, but political pressures have gradually eroded these controls. During droughts, grazing bans are often lifted to ease hardship – but rarely reinstated once conditions improve.

While a single nationwide grazing policy may be impractical, each state could develop its own, reflecting local culture, religious practices, and livestock needs while safeguarding the wildlife that depends on grasslands. Demonstration plots could help communities see how managed grazing improves fodder productivity and biodiversity alike.

### The Task Force and the Unfinished Agenda

Recognizing these challenges, the Planning Commission’s Environment and Forests Division, in August 2006, established a Task Force on Grasslands and Deserts to guide policy for the Eleventh Five-Year Plan (2007–2012). The Task Force emphasised that grasslands and deserts are the most neglected of India’s ecosystems, despite their immense role in supporting rural livelihoods and biodiversity.

With over 500 million livestock relying on them for fodder, and with many endangered species – from the great Indian bustard and lesser florican to the Nilgiri



Ramdevara Grassland near Pokharan





Painted Sandgrouse in Kolasa meadow of Tadoba-Andhari Tiger Reserve, Maharashtra



Chestnut-bellied Sandgrouse at the grasslands in Pokharan in Rajasthan

tahr and wild buffalo – depending on their survival, the Task Force called for urgent action. It recommended a National Grazing Policy, dedicated funding, and integrated research and conservation programmes. It urged the creation of a national network of grassland ecologists for long-term monitoring, and the establishment of dedicated divisions within the Ministry of Environment, Forest and Climate Change to oversee grassland conservation.

Sadly, as the report concluded, these recommendations received little more than a passing acknowledgement. The initiatives – Project Bustard, Project Snow Leopard, and others – remained largely to conducting population estimates, awaiting the conservation focus and resources they deserved.

After reviewing this account, BNHS wrote to the Union Minister, Mr Bhupender Singh Yadav, and to our surprise, we received a prompt response. He convened an urgent meeting and instructed officials to act quickly on several of the recommendations.

### A Landscape Waiting to Be Seen

India's grasslands continue to be termed as wastelands, though they are ancient ecosystems, shaped by millennia of natural cycles and human

coexistence. They feed livestock, shelter birds and mammals found nowhere else, and quietly store the carbon that regulate our climate. The challenge now is not simply to protect what remains, but to recognize the value of what has long been overlooked as wastelands.

For as the science has shown – from White in 2000 to the BNHS studies and the Task Force reports that followed – the story of the grasslands is not one of loss alone. It is also one of resilience. Given care, understanding, and policy that matches their ecological worth, these open lands can once again thrive, sustaining both people and the wild creatures that have shared them for centuries. 🌿



**Kishor Rithe**, Director, BNHS, has been working for wildlife conservation through sustainable livelihoods, conservation action, advocacy, and policy for over three decades.

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